Rabies Antibody Reconversion Following D1/D15/D30 Protocol in a Weimaraner with Primary FAVN Failure: Case Report and Comparative Analysis of Rabies Virus-Neutralizing Antibody Titres (KSVDL, 2026)

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Abstract

Background: The FAVN (Fluorescent Antibody Virus Neutralization) test is the internationally recognized standard for serological verification of rabies immune status in companion animals for international movement, with a regulatory threshold of ≥0.50 IU/mL adopted by WOAH, the European Union, and major national import frameworks (EU Regulation 576/2013; WOAH Chapter 8.14). Primary FAVN failures — results below threshold following an incomplete vaccination protocol or insufficient anamnestic response — represent a documented operational and clinical challenge.

Objective: To describe a case of primary FAVN failure in a 3-year-9-month male Weimaraner previously vaccinated under an undocumented protocol in Sweden, with subsequent successful serological reconversion under the Novibac Rabies D1/D15/D30 protocol at Veterinaria Zoovet Travel (Trujillo, Peru), and to present a comparative analysis of both KSVDL reports within the institutional FAVN data repository.

Case: Patient: ATON, Weimaraner, male, microchip 725093200213835. Initial KSVDL report R26-013079 (February 4, 2026): 0.22 IU/mL — FAIL. Prior vaccination: undocumented schedule administered in Sweden. Protocol applied: Novibac Rabies D1 (March 5, 2026), D15 (March 19, 2026), D30 blood draw (April 4, 2026). KSVDL report R26-030347 (April 4, 2026): ≥3.46 IU/mL — PASS.

Results: Post-protocol titre of ≥3.46 IU/mL represents a 15.73× delta above the initial FAIL value and 6.92× above the EU regulatory threshold. Within the active repository of 49 FAVN records maintained by Veterinaria Zoovet (2019–2026), this case is the only fully documented fail/reconversion pair and holds the highest reconversion delta on record.

Conclusion: The D1/D15/D30 Novibac Rabies protocol reversed primary serological failure in this case. Structured Dublin Core documentation of both KSVDL reports and integration into the institutional repository support clinical traceability and contribution to open-access datasets.

Keywords: FAVN; rabies virus-neutralizing antibodies; primary vaccine failure; serological reconversion; Novibac Rabies; D1/D15/D30; KSVDL; Weimaraner; international companion animal movement; Peru

1. Introduction

Rabies is a globally distributed zoonosis with a case fatality rate approaching 100% once clinical signs are established (WHO, 2023). Control at the human-animal interface depends on mass vaccination of the canine population — the principal reservoir and transmission vector worldwide — and on objective serological verification of that vaccination for international companion animal movement (Hampson et al., 2015).

Viral neutralization tests (VNT) — RFFIT and FAVN — are the international standard for quantifying rabies virus-neutralizing antibodies (rVNA). The ≥0.50 IU/mL threshold is universally recognized as an acceptable correlate of protection by WOAH, the European Union, and the United States (WOAH, 2023; Cliquet et al., 1998; CDC, 2024). The operational validity of this threshold depends, however, on the timing of sample collection relative to vaccination and on the vaccination schedule applied.

Primary FAVN failure — defined as a result below 0.50 IU/mL following rabies vaccination without adequate anamnestic response — has documented causes including: primary vaccination without a booster, premature sample collection (<30 days post-vaccination), interference from haemolysis or lipaemia, underlying immune deficiency, and use of vaccines with insufficient potency (McElhinney et al., 2026; Wallace et al., 2017). Undocumented vaccination schedules without a confirmed booster are consistently the most frequently reported cause of failure in accredited laboratory records.

The relevance of this case report is grounded in two converging contexts. First, the documented prevalence of imported dogs with sub-threshold rVNA titres: Belanger et al. (2025) found that approximately 48% of imported dogs had rVNA titres below 0.50 IU/mL at the time of arrival. Second, the scarcity of structured FAVN data from Latin America: at the time of writing, no other public repository of FAVN results from a Latin American veterinary operator is available in international repositories. The ATON case constitutes the only fully documented fail/reconversion pair in the active Veterinaria Zoovet dataset of 49 records (2019–2026), and holds the highest reconversion delta on record.

2. Case Description

2.1 Patient

ATON is an intact male Weimaraner, 3 years and 9 months of age at the time of the first sample collection (February 2026). ISO 11784/11785 microchip: 725093200213835. Owner resident in Trujillo, Peru, with final destination Sweden (rabies-free zone, European Union). No relevant clinical history at the time of consultation. Body weight and body condition score within normal parameters.

2.2 Prior Vaccination History

The owner reported that ATON had received a rabies vaccination during a stay in Sweden, administered by a veterinarian outside Veterinaria Zoovet Travel. At the time of consultation, it was not possible to verify: (a) the commercial vaccine brand used, (b) whether a one-dose or two-dose schedule had been applied, (c) the exact date of administration, or (d) whether a vaccination certificate or batch record existed. This absence of documentation made it impossible to determine whether the animal had received a primary vaccination with or without a booster, or whether the interval before sample collection met the minimum regulatory requirements (≥30 days post-primary vaccination; EU Reg. 576/2013).

KSVDL report R26-013079 (February 4, 2026) returned a result of 0.22 IU/mL, below the EU regulatory threshold of ≥0.50 IU/mL — equivalent to 44% of the minimum required value for international travel authorization to Sweden. This result determined the FAIL classification and referral to Veterinaria Zoovet Travel for application of the standard reconversion protocol.

2.3 Comparative KSVDL Reports

Parameter	R26-013079 (FAIL)	R26-030347 (PASS)
Animal / Microchip	ATON · 725093200213835	ATON · 725093200213835
Species / Breed / Sex / Age	Dog · Weimaraner · Male · 3Y 9M	Dog · Weimaraner · Male · 3Y 9M
Serum draw date	February 4, 2026	April 4, 2026
FAVN result (IU/mL)	0.22 IU/mL — FAIL	≥3.46 IU/mL — PASS
EU threshold (IU/mL)	< 0.50 (not met)	≥ 0.50 (6.92× threshold)
Laboratory	KSVDL — Kansas State	KSVDL — Kansas State
Vaccination protocol	Undocumented — Sweden	Novibac Rabies D1/D15/D30
Destination	Sweden (EU rabies-free)	Sweden (EU rabies-free)
Regulatory framework	EU Reg. 576/2013 · WOAH Ch. 8.14	EU Reg. 576/2013 · WOAH Ch. 8.14

Table 1. Comparative KSVDL parameters: R26-013079 (FAIL) vs. R26-030347 (PASS). Patient: ATON, Weimaraner, microchip 725093200213835. Veterinaria Zoovet Travel, Trujillo, Peru.

3. Reconversion Protocol D1/D15/D30

3.1 Immunological Basis

The D1/D15/D30 protocol with Novibac Rabies (MSD Animal Health) is based on the post-vaccination humoral immune response kinetics described for inactivated rabies vaccines in dogs. Following primary vaccination (D1), a lag phase of 7–10 days begins during which naive B lymphocytes process the antigen, proliferate, and begin differentiating into IgM-producing plasma cells. Class switching to IgG — the isotype with the greatest neutralizing capacity — occurs progressively over weeks 2–4. Administration of a booster on D15 — when memory B cells are fully primed but the primary response has not yet declined — generates a high-magnitude anamnestic response: rapid recruitment of memory plasmablasts, production of high-affinity IgG, and elevation of rVNA titres to levels detectable by FAVN at sample collection on D30, 16 days after the booster (Wallace et al., 2017; McElhinney et al., 2026).

The booster on D15 — day 15 of the protocol, 14 calendar days after D1 — rather than later has direct practical implications: it maximises the anamnestic response peak at the time of collection (D30), reduces the probability of a second FAVN FAIL that would require restarting the schedule, and optimises total process time for the owner.

3.2 Protocol Timeline

Figure 1. Novibac Rabies D1/D15/D30 reconversion protocol — ATON case. D1: primary vaccination March 5, 2026. D15: booster March 19, 2026 (day 15, 14 days elapsed). D30: blood draw April 4, 2026 (30 days elapsed post-D1). Veterinaria Zoovet Travel, Trujillo, Peru, 2026.

3.3 Regulatory Compliance

EU Regulation 576/2013 requires, for dogs from third countries not included in the list of authorised countries — a category that includes Peru — that the VNT sample (RFFIT or FAVN) be collected no earlier than 30 days after primary vaccination (European Commission, 2013). The D1/D15/D30 protocol meets this requirement exactly: the blood draw on April 4, 2026 (D30) is 30 calendar days after D1 (March 5, 2026), the minimum regulatory interval. The booster on D15 optimises the antibody titre without altering the permitted sampling window.

4. Results

4.1 FAVN R26-013079 — FAIL

Serum draw date: February 4, 2026. Laboratory: KSVDL (Kansas State Veterinary Diagnostic Laboratory), Manhattan, Kansas, USA — WOAH-accredited laboratory for FAVN testing. Result: 0.22 IU/mL. Classification: FAIL (<0.50 IU/mL). The result is equivalent to 44% of the minimum required value for international movement to Sweden (EU, rabies-free zone). Prior vaccination was reported as administered in Sweden under an undocumented schedule; no vaccine brand, administration date, or booster record was available.

4.2 FAVN R26-030347 — PASS

Serum draw date: April 4, 2026 (D30 of the D1/D15/D30 protocol). Laboratory: KSVDL. Result: ≥3.46 IU/mL. Classification: PASS (6.92× the regulatory threshold). The interval between the FAIL draw and the PASS draw was 59 days (February 4 – April 4, 2026), comprising the referral consultation, application of the D1/D15/D30 protocol, and KSVDL sample processing.

Reconversion delta: the post-protocol titre of ≥3.46 IU/mL represents: 15.73× the initial FAIL value (0.22 IU/mL → ≥3.46 IU/mL); 6.92× the EU regulatory threshold (0.50 IU/mL); and the highest reconversion delta recorded in the Veterinaria Zoovet FAVN repository (49 records, 2019–2026).

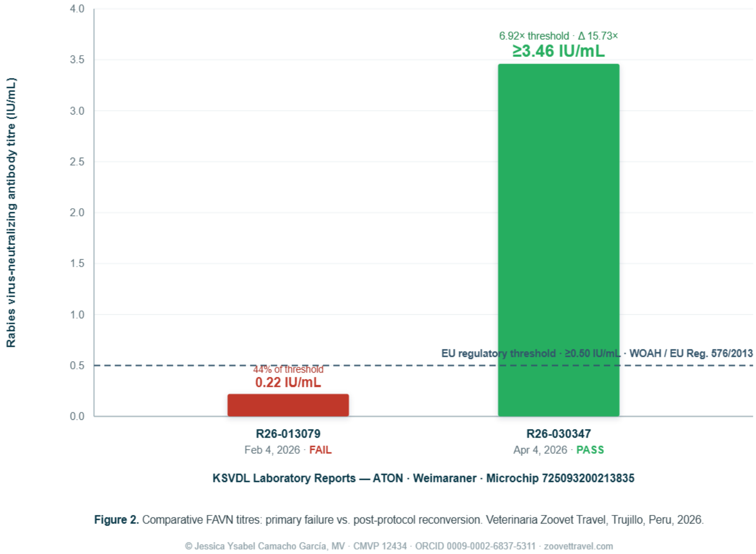

Figure 2. Comparative FAVN titres: primary failure vs. post-protocol reconversion. Veterinaria Zoovet Travel, Trujillo, Peru, 2026.

Figure 2. Comparative FAVN titres: primary failure vs. post-protocol reconversion. ≥3.46 IU/mL denotes a confirmed lower bound; actual titre may be higher. Patient: ATON, Weimaraner, microchip 725093200213835. KSVDL, 2026.

4.3 Dublin Core Cataloguing — Comparative

DC Element	R26-013079 (FAIL)	R26-030347 (PASS)
dc:title	FAVN Report — ATON · R26-013079 · FAIL	FAVN Report — ATON · R26-030347 · PASS
dc:creator	Jessica Ysabel Camacho García, MV	Jessica Ysabel Camacho García, MV
dc:subject	FAVN; rVNA; KSVDL; FAIL; subthreshold; reconversion candidate	FAVN; rVNA; KSVDL; PASS; post-booster reconversion
dc:date	2026-02-04	2026-04-04
dc:identifier	LAB# R26-013079 · KSVDL	LAB# R26-030347 · KSVDL
dc:source	Kansas State Veterinary Diagnostic Laboratory	Kansas State Veterinary Diagnostic Laboratory
dc:relation	retest: R26-030347	previous-fail: R26-013079
dc:coverage	Peru → Sweden · EU 576/2013 · Novibac D1/D15/D30	Peru → Sweden · EU 576/2013 · Novibac D1/D15/D30
dc:rights	© 2026 Jessica Ysabel Camacho García / Zoovet Travel	© 2026 Jessica Ysabel Camacho García / Zoovet Travel

Table 2. Dublin Core comparative cataloguing: R26-013079 (FAIL) vs. R26-030347 (PASS). Source: Veterinaria Zoovet Travel FAVN repository — index.json, records id:40 and id:41. Dataset DOI: https://doi.org/10.5281/zenodo.20311995.

4.4 Repository Context

The active Veterinaria Zoovet FAVN repository comprises 49 records (2019–2026) processed through KSVDL under the D1/D15/D30 Novibac Rabies protocol (for cases from 2024). Aggregate data: total records 49 (46 PASS / 3 FAIL); titre range 0.20–6.01 IU/mL; dataset mean 2.60 IU/mL; destinations include Spain (25), Italy (6), United States (5), Sweden (3), Germany, France, Portugal. The ATON pair (R26-013079 / R26-030347) is the only fully audited fail/reconversion pair in the dataset.

5. Discussion

The ATON case illustrates the operational consequence of undocumented vaccination schedules in the context of international companion animal movement to rabies-free zones. The prior vaccination administered in Sweden — with no known brand, no verifiable date, and no evidence of a booster — produced an rVNA titre of 0.22 IU/mL at the time of the KSVDL draw, a pattern consistent with a primary response without anamnestic amplification. This is in line with published evidence: Wallace et al. (2017) and McElhinney et al. (2026) identify the absence of a documented post-primary booster as the most frequent predictor of FAVN failure in dogs.

The reconversion delta of 15.73× is clinically meaningful. Published reference values for post-booster reconversion in dogs with prior FAVN failure are limited, and no comparable structured dataset from Latin American operators exists in international repositories at the time of writing. The ATON case therefore stands as an isolated but fully documented data point within the Zoovet Travel institutional dataset.

The ≥3.46 IU/mL value in the PASS report should be interpreted as a confirmed lower bound, not as an exact titre. The KSVDL notation indicates the result exceeds the upper quantification limit of the assay for that range; the actual titre may be higher. This does not affect the regulatory classification but is relevant when comparing across studies.

The prevalence of imported dogs with sub-threshold rVNA titres documented by Belanger et al. (2025) — approximately 48% below 0.50 IU/mL at arrival — suggests the scenario described here is not uncommon. The critical difference between this case and those arriving without adequate documentation is that ATON was tested serologically before any attempt to enter the destination country, as required by EU Regulation 576/2013 for third countries not on the authorised list. This allowed failure detection and protocol correction at origin. Regulatory frameworks that do not require pre-shipment serological testing create a documented biosecurity risk (CDC, 2024; Wallace et al., 2017).

From a data perspective, this case contributes the only audited fail/reconversion pair to the Veterinaria Zoovet dataset of 49 records. The first-draw PASS rate of 93.9% (46/49) supports the efficacy of the D1/D15/D30 protocol, and all three FAIL cases share a common antecedent: prior vaccination without a documented booster.

6. Limitations

The following limitations are noted. First, this is a single case; findings are not generalisable to the species or to other primary failure conditions. Second, prior vaccination history could not be verified documentally; an incomplete schedule, a low-potency vaccine, or a pre-threshold sample draw cannot be excluded as contributing factors. Third, the result ≥3.46 IU/mL is a confirmed lower bound, not an exact titre, and long-term functional immunity cannot be inferred from this report. Fourth, no haematological or biochemical data are available to exclude concurrent causes of immunosuppression.

7. Conclusion

The ATON case documents successful serological reconversion in a Weimaraner with primary FAVN failure (0.22 IU/mL, R26-013079, KSVDL) following the Novibac Rabies D1/D15/D30 protocol at Veterinaria Zoovet Travel, with a confirmed result of ≥3.46 IU/mL (R26-030347, KSVDL; delta 15.73×; 6.92× the EU threshold). Three operational conclusions follow: undocumented vaccination schedules produce insufficient primary responses detectable by FAVN; the requirement for verifiable vaccination documentation at origin is a serologically supported biosecurity measure; and the D1/D15/D30 Novibac Rabies protocol — with the booster timed within the immunological memory window on D15 — demonstrated efficacy in reversing primary failure in this case.

Structured Dublin Core documentation and integration into an open-access repository with a DOI are necessary conditions for contributing to the global evidence base on FAVN in Latin America.

Conflict of Interest Statement

The author declares that the D1/D15/D30 Novibac Rabies protocol described in this report is the standard protocol of Veterinaria Zoovet Travel, the author's affiliated institution. This may influence the interpretation and presentation of results. No external funding or financial conflict of interest is associated with this report.

Informed Consent

The patient's owner (ATON) authorized the use of clinical data and KSVDL laboratory results for preparation and publication of this case report. The owner's identity is not disclosed. Original records are available for editorial review upon request to the corresponding author.

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